AARON FOOTE

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EDUCATION

Wesleyan University, Middletown, CT — September 2020 - May 2025
M.A. in Computer Science — *GPA: 4.22/4.00*
B.A. with High Honors in Computer Science, Minor in Data Analysis — *GPA: 3.87/4.00*

WORK EXPERIENCE

Data Scientist — June 2025 - Present
CMC Foundation for Change — *Palo Alto, CA*

- Overhauling database with thousands of surveys on substance abuse therapy trainings with SPSS, R, and Python
- Developing data visualizations, reports, and dashboards for internal decisions and external fundraising/presentation
- Delivering internal presentations on key metrics and leading discussions on data-driven decision making
- Lead small research assistant team through data management, data analysis, and reporting

Data Engineer Intern — June 2025 - August 2025
Climecard — *Palo Alto, CA*

- Constructed SQLite database and RESTful APIs serving 100K+ properties nationwide for climate compliance tracking
- Leveraged AI to extract and synthesize previously inaccessible data from PDFs
- Built user-friendly emissions and energy dashboards for REIT client with $10 billion in assets

RESEARCH EXPERIENCE

Graduate Research Assistant – Interpretable ML — June 2024 - August 2025
Wesleyan University — *Middletown, CT*

- Developed a hypothesis test under minimal assumptions to evaluate feature importance methods for bias
- Validated the method through extensive simulation and application to real datasets in Python and R
- Integrated dimension reduction techniques to extend the test to higher dimension

Research Assistant – Sports Analytics — January 2024 - June 2024
Wesleyan University — *Middletown, CT*

- Produced an intuitive evaluation tool for NFL defenders using spatiotemporal player tracking data
- Leveraged ensemble ML algorithms to estimate player value and assign credit to defenders
- Designed and deployed an NCAA March Madness prediction model using Cox proportional hazards and bootstrapping
- Explored time-varying Cox model and random/deterministic bracket completion through extensive simulation

Research Assistant – Algorithms, Computational Complexity — September 2022 - May 2025
Wesleyan University — *Middletown, CT*

- Established complexity results and phase transition behavior for the solving of combinatorial puzzles
- Developed high-performance solvers in C/bash on HPC cluster
- Built web scraping pipeline and analyzed 25K+ puzzles published online with Python to identify difficulty heuristics

Summer Research Fellow – Computer Vision, Natural Language Processing — June 2021 - September 2021
Wesleyan University — *Middletown, CT*

- Developed a pipeline to extract text from political ads using corner detection and OCR, reducing costs by 100x
- Automated the workflow and post-processing with adaptive string clustering for over 200K ads

TEACHING EXPERIENCE

Substitute Teacher/Paraeducator — September 2025 - Present
Swing Education — *Palo Alto, CA*

- Foster a positive and inclusive learning environment that supported diverse student needs
- Deliver engaging lessons and maintained classroom management across multiple grade levels and subject areas
- Manage classroom behavior effectively by applying school policies and restorative practices
- Provide individualized academic assistance aligned with each student's IEP goals and accommodations

Head Quantitative Analysis Tutor — April 2023 - May 2025
Wesleyan University — *Middletown, CT*

- Held 10+ hours of weekly office hours open to the entire university, assisting with coursework, projects and research
- Provided one-on-one statistical consulting for undergraduate and graduate thesis projects
- Guided students in managing and analyzing data, employing and interpreting statistical techniques, and visualizing their work with R, STATA, Python, SPSS, SAS, and Excel
- Developed a schedule each semester, lead weekly meetings, and consult with professors

Teaching Assistant — September 2022 - December 2024
Wesleyan University — *Middletown, CT*

- Led weekly lab sections for 15+ students, explaining complex concepts through hands-on activities
- Graded 50+ assignments weekly on Moodle, providing detailed feedback to improve student understanding
- Courses: Algorithms, Theory of Computation, Survival Analysis, Applied Data Analysis

RESEARCH PRESENTATIONS

Foote, A. and Krizanc, D. (2025) "Nonogram: Complexity of Inference and Phase Transition Behavior." *arXiv Pre-Print* doi:10.48550/arXiv.2507.07283

Foote, A. and Krizanc, D. (2025) "TRIP: A Nonparametric Test to Diagnose Biased Feature Importance Scores." *IJCAI Workshop on Explainable Artificial Intelligence* (Oral Presentation) doi:10.48550/arXiv.2507.07276

Foote, A. and Matthews, G. (2024). "Continuous Within-Play Estimation of Run-Stopping Contribution in American Football." *Cascadia Symposium on Statistics in Sports* (Poster).

Foote, A., Gooyabadi, M. and Addleman, N. (2023). "Factors in Learning Dynamics Influencing Relative Strengths of Strategies in Poker Simulation." *Games.* doi:10.3390/g14060073.

Foote, A., Oleinikov, P. (2021). "Efficiently Extracting Text from Political Ad Videos." *CIS Summer Research Symposium.* (Poster)

AWARDS

Maureen Snow Prize (July 2023): Runner-up among 200 students, given for most effective communication of research to a broad audience

Baker '64 Collabria Fellowship in Data Analysis (April 2023) – **$5,000**

Best Data Story Award (April 2023): Awarded at Northeast Division ASA DataFest Competition

QAC Apprentice Grant (April 2021) – **$5,400**

TECHNICAL SKILLS

Tools and Languages: R, Python (NumPy, Pandas, TensorFlow, matplotlib, scikit-learn), SPSS, STATA, SAS, Excel, SQL, C, Typescript

Methods: neural networks, statistical learning, linear regression, data visualization, exploratory data analysis, hypothesis testing, time series, survival analysis, data engineering